

GAP-10T: Algebraic Torsion Selection in the Minimal First-Order Branch

Ing. David Jaroš

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Abstract

The kinematic identity $\omega = \dot{\omega}(e) + K(T)$ does not determine T . This note closes a sharply defined dynamical subgap. In the minimal Hilbert–Palatini/Einstein–Cartan branch, with nondegenerate tetrad and no independent torsion kinetic term, variation of the Lorentz connection gives an algebraic Cartan equation. Its torsion map is pointwise invertible: vanishing spin current forces $T = 0$, while a specified spin current determines a unique contorsion. For a two-sided UBT matter derivative the spin current is obtained by varying the same paired connection acting on Θ . What remains open is not the algebra of torsion elimination, but the derivation of this minimal first-order action, its exact UBT representation and normalization, and the resulting Θ spin current from the fundamental theory.

1 First-order branch

Let e^a be a nondegenerate tetrad, $\omega^{ab} = -\omega^{ba}$ an independent Lorentz connection, and

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}, \quad T^a = de^a + \omega^a{}_b \wedge e^b.$$

Consider the minimal first-order action

$$S_{\text{HP}}[e, \omega] = \frac{1}{4\kappa} \int \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\omega) - \frac{\Lambda}{24\kappa} \int \epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \quad (1)$$

plus a matter/UBT action $S_{\text{m}}[e, \omega, \Theta]$. Define the Lorentz-spin current three-form by the convention

$$\delta_\omega S_{\text{m}} = \frac{1}{2} \int \tau_{ab} \wedge \delta\omega^{ab}. \quad (2)$$

2 Connection variation

Theorem 2.1 (Cartan connection equation). *For compactly supported variations of ω , the connection Euler–Lagrange equation of Eq. (1) is*

$$\boxed{\epsilon_{abcd} T^a \wedge e^b = \kappa \tau_{cd}.} \quad (3)$$

The equation is algebraic in torsion: it contains no derivative of T .

Proof. Use $\delta R^{cd} = D_\omega \delta \omega^{cd}$ and integrate by parts. Since $D_\omega(e^a \wedge e^b) = T^a \wedge e^b - e^a \wedge T^b$, antisymmetry of ϵ_{abcd} combines the two terms into $2\epsilon_{abcd}T^a \wedge e^b$. Adding Eq. (2) gives Eq. (3). The cosmological term is independent of ω . $\square$

Theorem 2.2 (Pointwise invertibility of the Cartan torsion map). *For a nondegenerate tetrad in four dimensions, the linear map*

$$\mathcal{C}_e : T^a \longmapsto \epsilon_{abcd} T^a \wedge e^b \quad (4)$$

from the 24 torsion components to the 24 Lorentz-valued three-form components is an isomorphism.

Proof. The claim is pointwise and Lorentz covariant, so choose an orthonormal frame $e^a = dx^a$. Expanding the six antisymmetric pairs (c, d) in the four three-form basis elements gives a 24×24 linear system. Direct row reduction has rank 24; the exact verifier is `tools/verify_remaining_gr_subclosures.py`. Hence the kernel is zero in one orthonormal frame and therefore in every nondegenerate frame. $\square$

Corollary 2.3 (Vacuum/minimal bosonic torsion). *If the matter action has zero Lorentz-spin current, $\tau_{ab} = 0$, then Eq. (3) uniquely gives*

$$\boxed{T^a = 0, \quad K^{ab} = 0, \quad \omega^{ab} = \dot{\omega}^{ab}(e).} \quad (5)$$

Corollary 2.4 (Spin-sourced torsion). *For a specified τ_{ab} , Eq. (3) determines a unique torsion and hence a unique contorsion $K(T)$. In the minimal branch torsion is nonpropagating: it is an auxiliary field algebraically tied to spin current.*

3 Two-sided UBT spin current

For the generic algebra-native derivative

$$D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu, \quad (6)$$

a Lorentz variation acts through a paired generator

$$\mathcal{J}_{ab} \Theta = J_{ab}^L \Theta - \Theta J_{ab}^R. \quad (7)$$

If the kinetic term uses an invariant real bilinear $\langle \cdot, \cdot \rangle$, variation of the connection produces a current schematically of the form

$$\tau_{ab} \propto * \text{Sym} \langle D\Theta, \mathcal{J}_{ab} \Theta \rangle, \quad (8)$$

with the exact sign and normalization fixed by the canonical action and its left/right representation. Equation (8) is not set to zero by calling Θ “bosonic”: a field carrying a nontrivial Lorentz representation can source the independent spin connection.

4 What is closed and what remains

Defensible status

GAP-10T-PALATINI: CLOSED CONDITIONALLY [L1]. For the minimal Hilbert–Palatini branch with nondegenerate tetrad and no independent torsion kinetic term, the connection equation is algebraic and pointwise invertible. Zero spin current gives zero torsion; specified spin current gives unique contorsion.

GAP-10T-DYN: NARROWED, NOT FULLY CLOSED. The remaining UBT task is to derive the first-order gravitational term and the precise paired matter representation from the canonical action, calculate its spin current and normalization, and decide whether additional torsion-squared or propagating-torsion invariants are present.

The standard first-order geometry used here is not claimed as a new result. The UBT-specific gain is the exact separation between (i) the already closed kinematic reconstruction $\omega(e, T)$, (ii) the algebraically closed minimal torsion equation, and (iii) the still-open origin and field content of the canonical UBT action.

References

- [1] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, “General Relativity with Spin and Torsion: Foundations and Prospects,” *Rev. Mod. Phys.* **48**, 393–416 (1976).
- [2] R. M. Wald, *General Relativity*, University of Chicago Press (1984).